

EJERCICIOS BLOQUE 3 nivel básico

Ejercicios de red (1–12)

◆ Bloque 1: Información del sistema y red (1–8)

Ejercicio 1. Información del sistema

Muestra el nombre del equipo, usuario actual y sistema operativo.

Get-ComputerInfo | Select-Object CsName, WindowsVersion, OsArchitecture

```
1 clear
2 Get-ComputerInfo | Select-Object CsName, WindowsVersion, OsArchitecture
3
```

CsName	WindowsVersion	OsArchitecture
JUANCARLOSD905	2009	64-bit

PS C:\WINDOWS\system32>

Ejercicio 2. Adaptadores de red

Lista los adaptadores de red mostrando nombre y estado.

Get-NetAdapter | Select-Object Name, Status

```
1 clear
2 Get-NetAdapter | Select-Object Name, Status
3
4
```

Name	Status
Ethernet	Up

PS C:\WINDOWS\system32>

Ejercicio 3. Dirección IP local

Muestra la dirección IPv4 principal del equipo.

```
Get-NetIPAddress -AddressFamily IPv4 | Where-Object {$_.IPAddress -notlike "169.*"} |  
Select-Object InterfaceAlias, IPAddress
```

```
clear  
Get-NetIPAddress -AddressFamily IPv4 | Where-Object {$_.IPAddress -notlike "169.*"} | Select-Object InterfaceAlias, IPAddress
```

InterfaceAlias	IPAddress
Ethernet	192.168.0.128
Loopback Pseudo-Interface 1	127.0.0.1

PS C:\WINDOWS\system32> |

Ejercicio 4. Ping básico

Envía 4 pings a google.com.

```
Test-Connection google.com -Count 4
```

```
1 clear  
2 Test-Connection google.com -Count 4  
3  
4
```

Source	Destination	IPv4Address	IPv6Address	Bytes	Time(ms)
JUANCARLOS...	google.com	142.250.178.174		32	15
JUANCARLOS...	google.com	142.250.178.174		32	15
JUANCARLOS...	google.com	142.250.178.174		32	15
JUANCARLOS...	google.com	142.250.178.174		32	15

Ejercicio 5. Ping a varios hosts

Haz ping a google.com, microsoft.com y github.com.

```
$hosts = "google.com","microsoft.com","github.com"
```

```
foreach ($h in $hosts) { Test-Connection $h -Count 2 | Select-Object Address,  
ResponseTime }
```

```
1 clear  
2 $hosts = "google.es", "larazon.com", "marca.com"  
3 foreach ($h in $hosts){  
4     Test-Connection $h -Count 2 | Select-Object Address, ResponseTime  
5 }
```

Address	ResponseTime
google.es	15
google.es	14
larazo...	228
larazo...	229
marca.com	49
marca.com	49

Ejercicio 6. Comprobar puerto abierto

Comprueba si www.microsoft.com responde en el puerto 80.

Test-NetConnection -ComputerName www.microsoft.com -Port 80

```
1 clear
2 Test-NetConnection -ComputerName microsoft.com -Port 80
```



```
ComputerName      : microsoft.com
RemoteAddress     : 13.107.246.43
RemotePort        : 80
InterfaceAlias    : Ethernet
SourceAddress     : 192.168.0.128
TcpTestSucceeded  : True
```

Ejercicio 7. Ruta hacia un servidor (tracert)

Muestra la ruta hasta 8.8.8.8.

Test-NetConnection 8.8.8.8 -TraceRoute

```
1 clear
2 Test-NetConnection 8.8.8.8 -TraceRoute
```

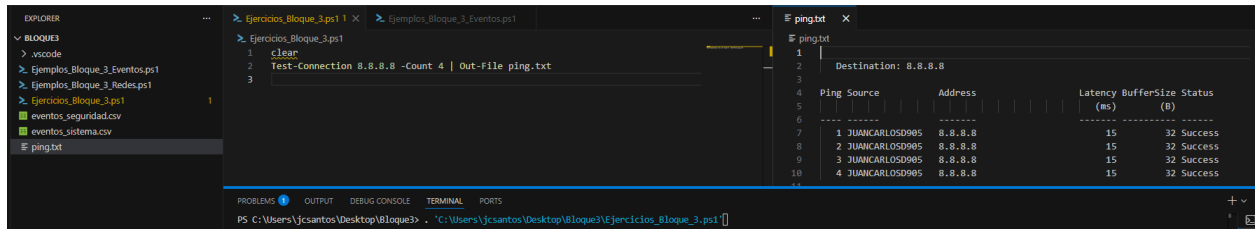


```
ComputerName      : 8.8.8.8
RemoteAddress     : 8.8.8.8
InterfaceAlias    : Ethernet
SourceAddress     : 192.168.0.128
PingSucceeded     : True
PingReplyDetails (RTT) : 15 ms
TraceRoute        : 192.168.0.1
                  10.187.4.161
                  10.143.80.4
                  81.45.103.33
                  0.0.0.0
                  81.46.0.17
                  176.52.253.93
                  213.140.50.43
                  108.170.249.11
                  74.125.253.203
                  8.8.8.8
```

Ejercicio 8. Guardar resultados de ping

Haz ping a 8.8.8.8 y guarda el resultado en un archivo de texto.

Test-Connection 8.8.8.8 -Count 4 | Out-File "\$env:USERPROFILE\Desktop\ping.txt"



The screenshot shows the VS Code interface. On the left, the Explorer pane shows a file named 'ping.txt'. The main editor shows a PowerShell script in 'Ejercicios_Bloque_3.ps1' with the following content:

```
1 clear
2 Test-Connection 8.8.8.8 -Count 4 | Out-File ping.txt
3
```

On the right, the 'ping.txt' file is open, displaying the results of the ping command:

```
1
2 Destination: 8.8.8.8
3
4 Ping Source Address Latency BufferSize Status
5 (ms) (B)
6 -----
7 1 JUANCARLOS0905 8.8.8.8 15 32 Success
8 2 JUANCARLOS0905 8.8.8.8 15 32 Success
9 3 JUANCARLOS0905 8.8.8.8 15 32 Success
10 4 JUANCARLOS0905 8.8.8.8 15 32 Success
```

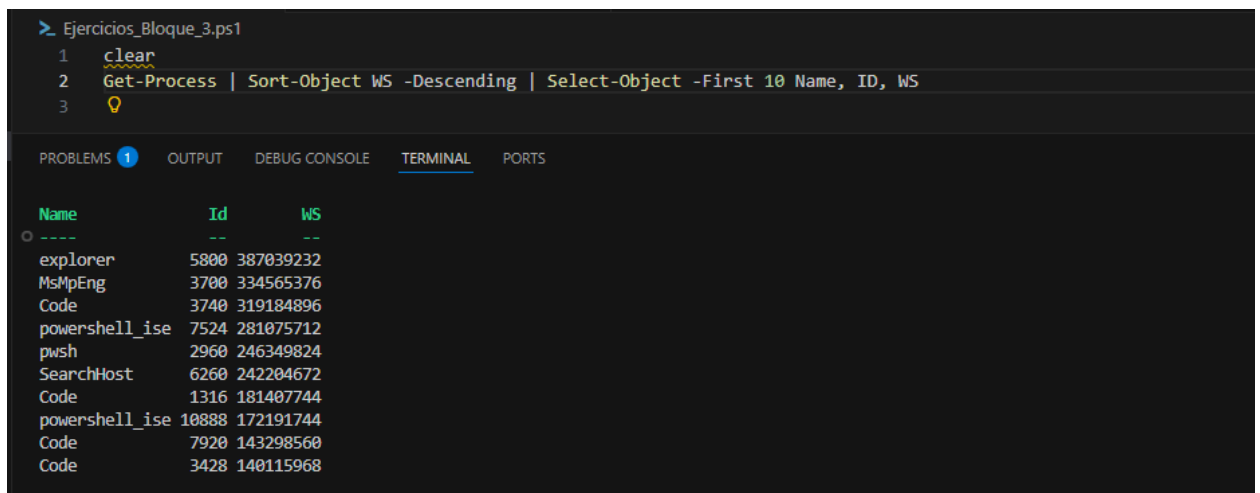
The bottom status bar shows the terminal path: 'PS C:\Users\jcsantos\Desktop\Bloque3> . "C:\Users\jcsantos\Desktop\Bloque3\Ejercicios_Bloque_3.ps1"'.

◆ Bloque 2: Exploración del sistema y archivos (9–12)

Ejercicio 9. Listar procesos activos

Muestra los 10 primeros procesos con más uso de memoria.

Get-Process | Sort-Object WS -Descending | Select-Object -First 10 Name, ID, WS



The screenshot shows the VS Code interface. The main editor shows a PowerShell script in 'Ejercicios_Bloque_3.ps1' with the following content:

```
1 clear
2 Get-Process | Sort-Object WS -Descending | Select-Object -First 10 Name, ID, WS
3
```

The bottom terminal pane shows the output of the script:

Name	Id	WS
explorer	5800	387039232
MsMpEng	3700	334565376
Code	3740	319184896
powershell_ise	7524	281075712
pwsh	2960	246349824
SearchHost	6260	242204672
Code	1316	181407744
powershell_ise	10888	172191744
Code	7920	143298560
Code	3428	140115968

Ejercicio 10. Servicios en ejecución

Muestra los servicios que están en estado "Running".

Get-Service | Where-Object Status -eq "Running" | Select-Object Name, DisplayName

```
1 clear
2 Get-Service | Where-Object Status -eq "Running" | Select-Object Name, DisplayName, Status
3
```

Name	DisplayName	Status	ID
AppIDSvc	Application Identity	Running	--
Appinfo	Application Information	Running	
AudioEndpointBuilder	Windows Audio Endpoint Builder	Running	
Audiosrv	Windows Audio	Running	
BFE	Base Filtering Engine	Running	
BrokerInfrastructure	Background Tasks Infrastructure Service	Running	
camsvc	Capability Access Manager Service	Running	

Ejercicio 11. Espacio en disco

Muestra las unidades disponibles y el espacio libre.

Get-PSDrive -PSProvider FileSystem | Select-Object Name, Used, Free

```
1 clear
2 Get-PSDrive -PSProvider FileSystem | Select-Object Name,
3 @{Name='UsedGB';Expression={[math]::Round($_.Used / 1GB, 2)}},
4 @{Name='FreeGB';Expression={[math]::Round($_.Free / 1GB, 2)}}
```

Name	UsedGB	FreeGB
C	33.84	220.81
D	0	0

PS C:\WINDOWS\system32>

Ejercicio 12. Listar archivos de una carpeta

Pide al usuario una ruta y muestra los archivos.

\$ruta = Read-Host "Introduce una ruta"

Get-ChildItem \$ruta

```
1 clear
2 $directorio = Read-Host "Dime el directorio a mostrar"
3 Get-ChildItem $directorio
```

Dime el directorio a mostrar: c:/Mac

Directory: C:\Mac

Mode	LastWriteTime	Length	Name
d----	01/29/2026 12:12 p. m.		Home
-a----	01/29/2026 12:12 p. m.	71	ReadMe.ur1

PS C:\WINDOWS\system32>

◆ Bloque 3: Eventos del sistema (13–16)

Ejercicio 13. Últimos 10 eventos del sistema

Get-WinEvent -LogName System -MaxEvents 10 | Select-Object TimeCreated, Id, LevelDisplayName, ProviderName

```
Ejercicios_Bloque_3.ps1
1 clear
2 Get-WinEvent -LogName System -MaxEvents 10 | Select-Object TimeCreated, Id, LevelDisplayName, ProviderName
3
```

TimeCreated	Id	LevelDisplayName	ProviderName
02/27/2026 11:16:42 a. m.	1	Information	Microsoft-Windows-Kernel-General
02/27/2026 11:16:42 a. m.	24	Information	Microsoft-Windows-Kernel-General
02/27/2026 11:05:10 a. m.	1	Information	Microsoft-Windows-Kernel-General
02/27/2026 11:05:10 a. m.	24	Information	Microsoft-Windows-Kernel-General
02/27/2026 10:08:13 a. m.	16	Information	Microsoft-Windows-Kernel-General
02/27/2026 9:20:43 a. m.	16	Information	Microsoft-Windows-Kernel-General
02/27/2026 9:20:43 a. m.	16	Information	Microsoft-Windows-Kernel-General
02/27/2026 9:20:09 a. m.	19	Information	Microsoft-Windows-WindowsUpdateClient
02/27/2026 9:20:06 a. m.	43	Information	Microsoft-Windows-WindowsUpdateClient
02/27/2026 9:20:05 a. m.	16	Information	Microsoft-Windows-Kernel-General

PS C:\Users\jcsantos\Desktop\Bloque3> . 'C:\Users\jcsantos\Desktop\Bloque3\Ejercicios_Bloque_3.ps1'

Ejercicio 14. Últimos errores de aplicación

Get-WinEvent -FilterHashtable @{LogName='Application'; Level=2} -MaxEvents 5

```
Ejercicios_Bloque_3.ps1
1 clear
2 Get-WinEvent -FilterHashtable @{LogName='Application'; Level=2} -MaxEvents 5
3
```

ProviderName: Firefox Default Browser Agent

TimeCreated	Id	LevelDisplayName	Message
02/27/2026 8:13:16 a. m.	5		

ProviderName: Microsoft-Windows-Security-SPP

TimeCreated	Id	LevelDisplayName	Message
02/26/2026 8:53:41 a. m.	8198	Error	License Activation (slui.exe) failed with the following error code:...

ProviderName: MariaDB

TimeCreated	Id	LevelDisplayName	Message
01/30/2026 8:41:40 a. m.	100	Error	mysqld.exe: Aria engine: checkpoint failed.
01/30/2026 8:41:40 a. m.	100	Error	mysqld.exe: Can't sync file 'C:\xampp\mysql\data\aria_log.00000001' to disk (Errcode: 9 "Bad file descriptor")...
01/30/2026 8:41:40 a. m.	100	Error	mysqld.exe: Error writing file 'C:\xampp\mysql\data\aria_log.00000001' (Errcode: 9 "Bad file descriptor")...

PS C:\Users\jcsantos\Desktop\Bloque3> . 'C:\Users\jcsantos\Desktop\Bloque3\Ejercicios_Bloque_3.ps1'

Ejercicio 15. Contar eventos recientes

(Get-WinEvent -LogName System -MaxEvents 50).Count

```
Ejercicios_Bloque_3.ps1
1 clear
2 (Get-WinEvent -LogName System).Count
3
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

1446

PS C:\Users\jcsantos\Desktop\Bloque3> . 'C:\Users\jcsantos\Desktop\Bloque3\Ejercicios_Bloque_3.ps1'

Ejercicio 16. Exportar eventos a CSV

Get-WinEvent -LogName System -MaxEvents 20 | Export-Csv
"\$env:USERPROFILE\Desktop\eventos.csv" -NoTypeInfoInformation

```
Ejercicios_Bloque_3.ps1
1 clear
2 Get-WinEvent -LogName System | Export-Csv all_eventos.csv -NoTypeInfoInformation
3
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\jcsantos\Desktop\Bloque3> . 'C:\Users\jcsantos\Desktop\Bloque3\Ejercicios_Bloque_3.ps1'

all_eventos.csv > data

```
1 "Message","Id","Version","Qualifiers","Level","Task","Opcode","Keywords","RecordId"
2 "The system time has changed to [U+200E]2026[U+200E] - [U+200E]02[U+200E] - [U+200E]
3 Time Delta: 391814 ms
4
5 Change Reason: An application or system component changed the time.
6 Process: '\Device\HarddiskVolume4\Program Files\Parallels\Parallels Tools\Serv
7
8 RTC time: [U+200E]2026[U+200E] - [U+200E]02[U+200E] - [U+200E]27T11:16:42.723000000
9 Current time zone b1 Col 1: Message
10 RTC time is in UTC: false
11 System time was based on RTC time: false,"1","4","4","5","0","-92233720368547
12 "The time zone information was refreshed with exit reason 0. Current time zone
13 "The system time has changed to [U+200E]2026[U+200E] - [U+200E]02[U+200E] - [U+200E]
14 Time Delta: 1796927 ms
15
16 Change Reason: An application or system component changed the time.
17 Process: '\Device\HarddiskVolume4\Program Files\Parallels\Parallels Tools\Serv
18
19 RTC time: [U+200E]2026[U+200E] - [U+200E]02[U+200E] - [U+200E]27T11:05:10.827000000
20 Current time zone bias: -60
21 RTC time is in UTC: false
22 System time was based on RTC time: false,"1","4","4","5","0","-92233720368547
23 "The time zone information was refreshed with exit reason 0. Current time zone
24 "The access history in hive \??\C:\Users\Default\NTUSER.DAT was cleared updatin
25 "The access history in hive \??\C:\Users\jcsantos\AppData\Local\Packages\Wicr
```

◆ Bloque 4: Tareas y automatización (17–20)

Ejercicio 17. Listar tareas programadas

Get-ScheduledTask | Select-Object TaskName, State

```
1 clear
2 Get-ScheduledTask | Select-Object TaskName, State
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

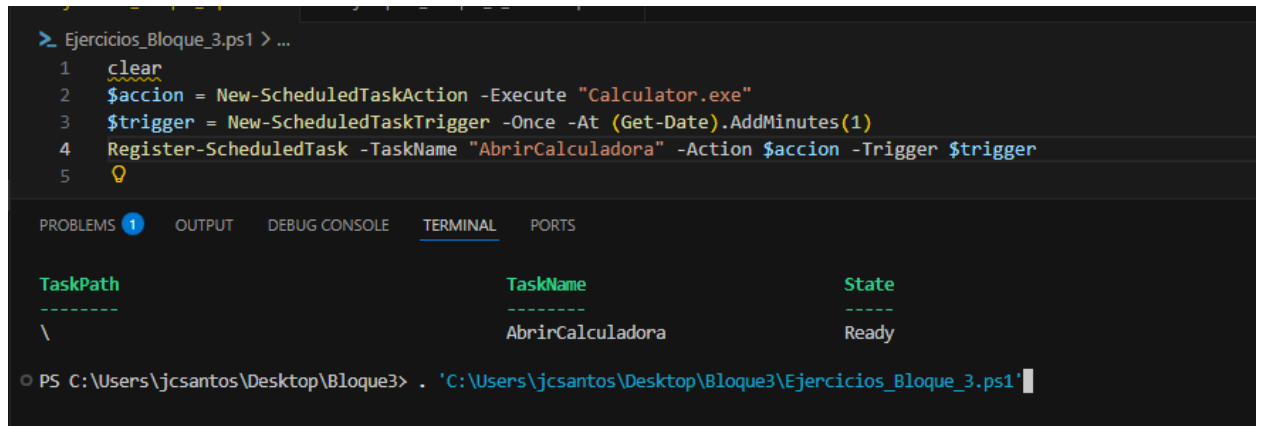
TaskName	State
C:\Program Files\HeidiSQL\heidisql.exe	Ready
MicrosoftEdgeUpdateTaskMachineCore{7B94667B-5427-4680-9577-F9A3412C281A}	Running
MicrosoftEdgeUpdateTaskMachineUA{3AE6BC93-DAA6-45B4-9571-1F6AA9089F2C}	Ready
.NET Framework NGEN v4.0.30319	Ready
.NET Framework NGEN v4.0.30319 64	Ready
.NET Framework NGEN v4.0.30319 64 Critical	Disabled
.NET Framework NGEN v4.0.30319 Critical	Disabled
AD RMS Rights Policy Template Management (Automated)	Disabled
AD RMS Rights Policy Template Management (Manual)	Ready
EDP Policy Manager	Ready
PolicyConverter	Ready
VerifiedPublisherCertStoreCheck	Ready
MareBackup	Ready
Microsoft Compatibility Appraiser	Disabled

Ejercicio 18. Crear tarea para abrir el bloc de notas

```
$saccion = New-ScheduledTaskAction -Execute "notepad.exe"
```

```
$strigger = New-ScheduledTaskTrigger -Once -At (Get-Date).AddMinutes(1)
```

```
Register-ScheduledTask -TaskName "AbrirNotepad" -Action $saccion -Trigger $strigger
```



```
> Ejercicios_Bloque_3.ps1 > ...
1 clear
2 $saccion = New-ScheduledTaskAction -Execute "Calculator.exe"
3 $strigger = New-ScheduledTaskTrigger -Once -At (Get-Date).AddMinutes(1)
4 Register-ScheduledTask -TaskName "AbrirCalculadora" -Action $saccion -Trigger $strigger
5
```

TaskPath	TaskName	State
\	AbrirCalculadora	Ready

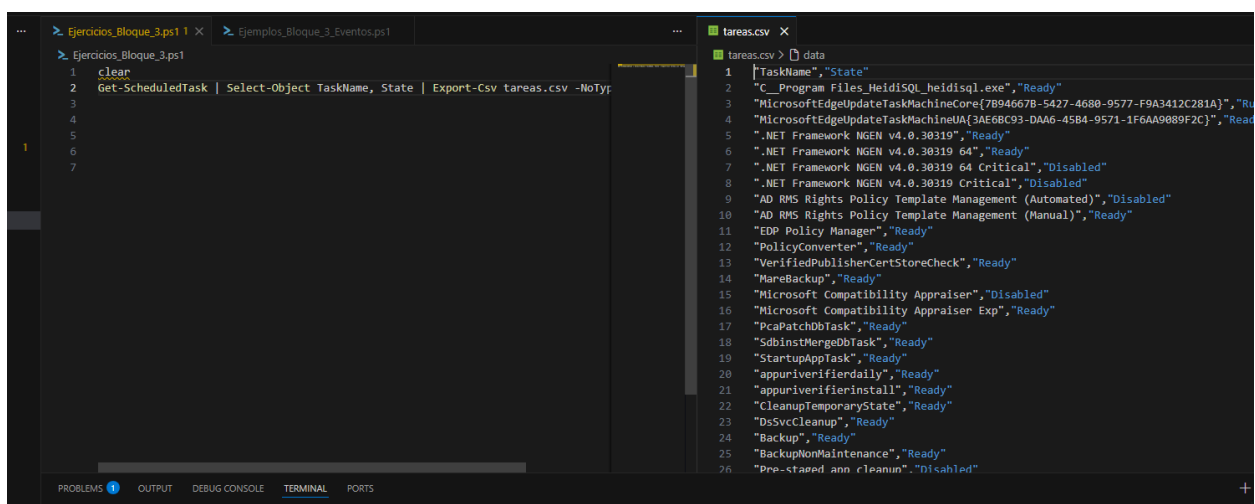
PS C:\Users\jcsantos\Desktop\Bloque3> . 'C:\Users\jcsantos\Desktop\Bloque3\Ejercicios_Bloque_3.ps1'

Ejercicio 19. Ejecutar manualmente una tarea

```
Start-ScheduledTask -TaskName "AbrirNotepad"
```

Ejercicio 20. Exportar tareas a CSV

```
Get-ScheduledTask | Select-Object TaskName, State | Export-Csv  
"$env:USERPROFILE\Desktop\tareas.csv" -No
```



```
> Ejercicios_Bloque_3.ps1 X Ejemplos_Bloque_3_Eventos.ps1
1 clear
2 Get-ScheduledTask | Select-Object TaskName, State | Export-Csv tareas.csv -NoTypeInformation
3
4
5
6
7
```

TaskName	State
"C:\Program Files\HeidiSQL\heidisql.exe"	"Ready"
"MicrosoftEdgeUpdateTaskMachineCore(7B94667B-5427-4680-9577-F9A3412C281A)"	"Ready"
"MicrosoftEdgeUpdateTaskMachineUA(3AE68C93-DAA6-45B4-9571-1F6AA989F2C)"	"Ready"
".NET Framework NGEN v4.0.30319"	"Ready"
".NET Framework NGEN v4.0.30319 64"	"Ready"
".NET Framework NGEN v4.0.30319 64 Critical"	"Disabled"
".NET Framework NGEN v4.0.30319 Critical"	"Disabled"
"AD RMS Rights Policy Template Management (Automated)"	"Disabled"
"AD RMS Rights Policy Template Management (Manual)"	"Ready"
"EDP Policy Manager"	"Ready"
"PolicyConverter"	"Ready"
"VerifiedPublisherCertStoreCheck"	"Ready"
"WareBackup"	"Ready"
"Microsoft Compatibility Appraiser"	"Disabled"
"Microsoft Compatibility Appraiser Exp"	"Ready"
"PcaPatchDbTask"	"Ready"
"SdbinstMergeDbTask"	"Ready"
"StartupAppTask"	"Ready"
"appuriverifierdaily"	"Ready"
"appuriverifierinstall"	"Ready"
"CleanupTemporaryState"	"Ready"
"DssvcCleanup"	"Ready"
"Backup"	"Ready"
"BackupNonMaintenance"	"Ready"
"Pre-staged ann cleanup"	"Disabled"